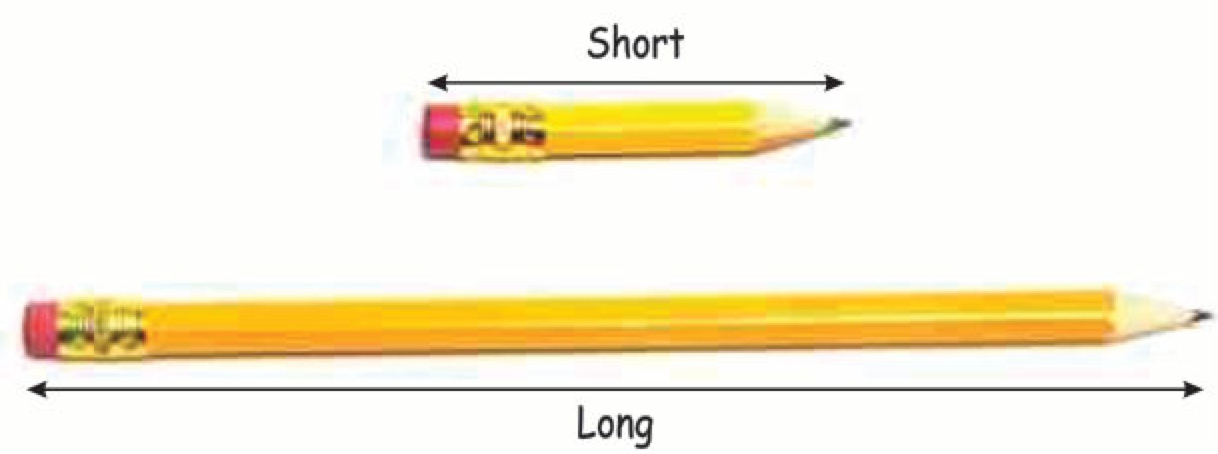
In previous class, we have measured, compared and ordered the length, mass and capacity of different objects using non-standard units. Now, we will use standard units to measure, compare and order the length, mass and capacity of different objects.

Length:

Every object has its length. A length tells us that whether an object is long or short.



In the above figure, the long pencil has the larger length than the short pencil.

Standard units of length (meter and centimeter):

Do you know how long a pencil is? Or how long your ruler is? We use units like meters and centimeters to measure the length of things just like how we use numbers to count. We use these units to measure how long something is.

What is a Meter?

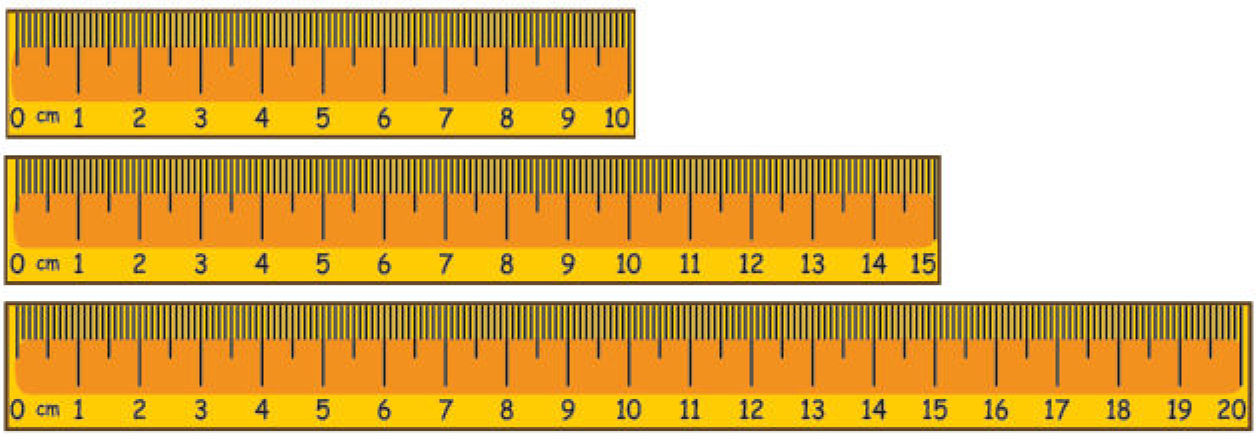
A meter (**m**) is a standard unit of length. We use meters to measure larger things like trains, streets and even jogging track.

In the above figure, the meter tape which is used to measure the length of different objects.

The above figure represents the meter rode of length 1 unit.

What is a Centimeter?

A centimeter (**cm**) is a smaller unit of length. We use centimeters to measure the length of smaller things like pencils, books, and even our own height.



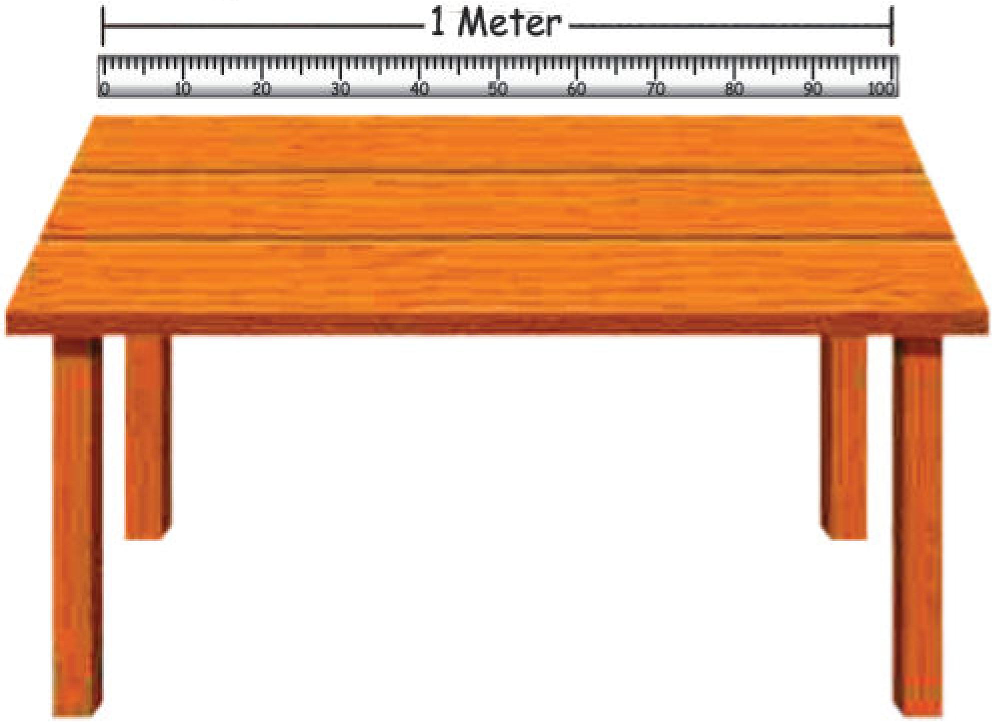
Above figures represent the ruler of length 10cm, 15cm, 20cm respectively.

Measuring with Meters and Centimeters:

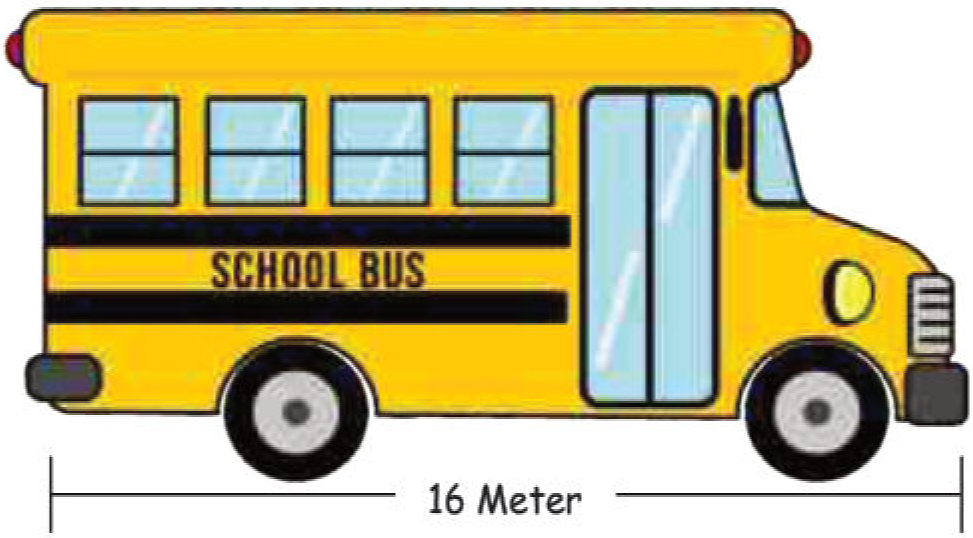
We can use meters and centimeters to measure the length of different objects. We can measure how long a book is in centimeters, or how long a bus is in meters!

Example:

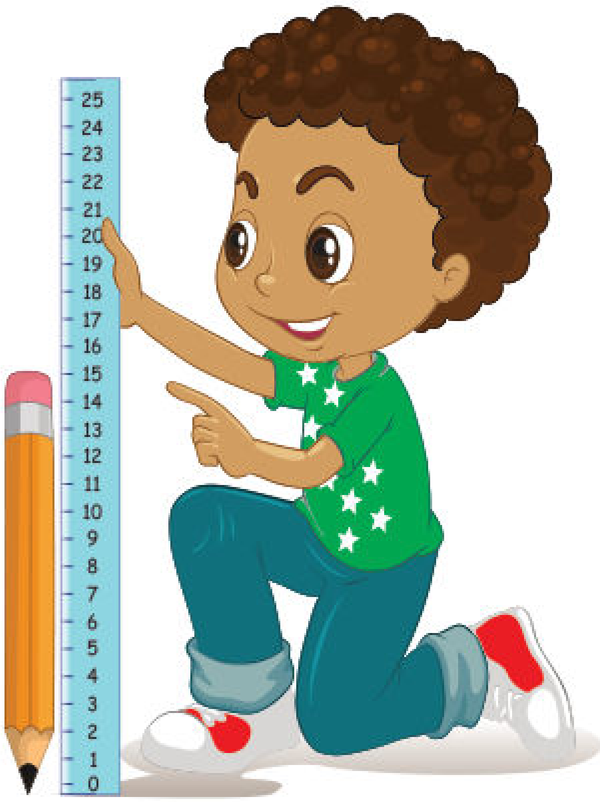
- A table of length 1 meter.



- A bus is about 16 meter long.



- A pencil is about 15 centimeters long.



- A book is about 20 centimeters long.

Activity

Write the measurement of the following objects

(i) Length of a  is measured in

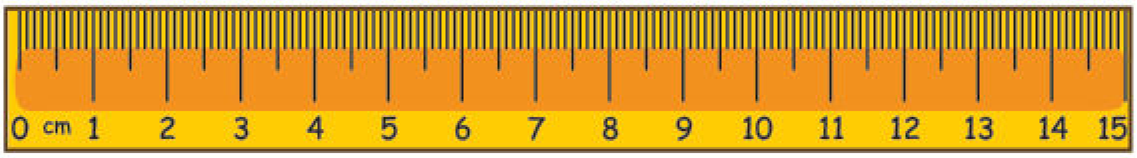
(ii) Length of a  is measured in

(iii) Your height  is measured in

(iv) Height of a  is measured in

Exercise 1

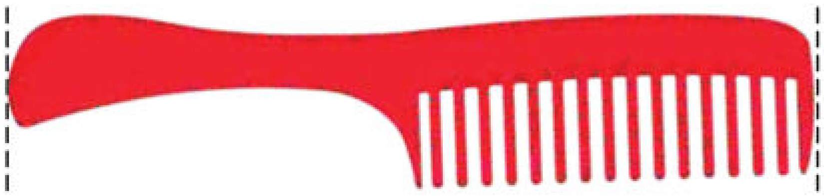
1. Measure the length of objects and write in centimeters:

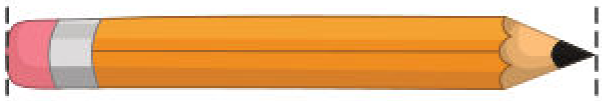


(i) 

(ii) 

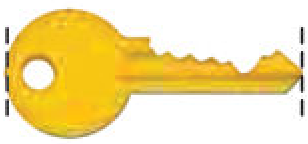
(iii) 

(iv) 

(v) 

(vi) 

(vii) 

(viii) 

2. Measure the length of objects in your classroom and write it in meters. (Rounded in units)

   1. Length of Table in the classroom =----------------

   2. Length of door in the classroom =----------------

   3. Length of black board in the classroom =----------------

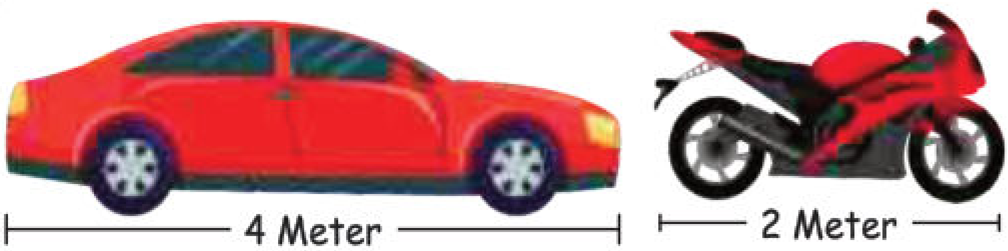
   4. Length of desk in the classroom =----------------

Comparison of length of different objects:

We can compare the length of different objects using meters and centimeters. We can say if one object is longer or shorter than another object.

Example 1:

Compare the length of a car (4 meter) with that of a bike (2 meter).

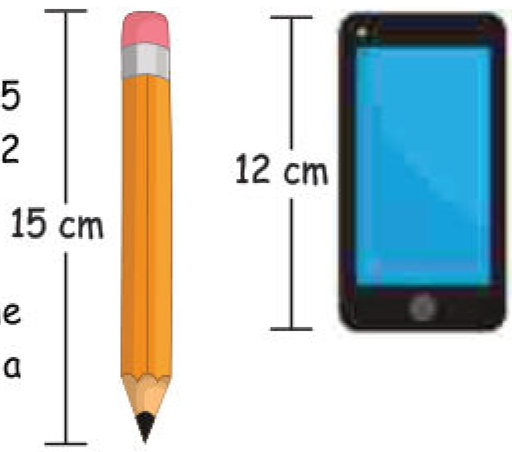


As, 4 is greater than 2. So, the length of a car is **greater** than that of a bike.

Example 2:

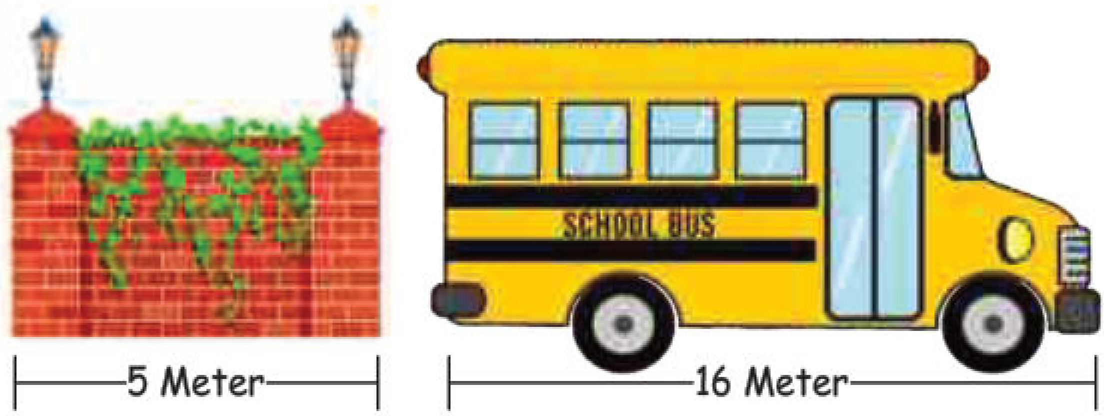
Compare the length of a pencil (15 centimeter) with that of a mobile (12 centimeter).

As, 15 is **greater** than 12. So, the length of pencil is greater than that of a mobile.



Example 3:

Compare the length of a wall (5 meter) with that of a bus (16 meter).



As, 5 is less than 16. So, the length of the wall (5 meter) is **less** than that of a bus (16 meter).

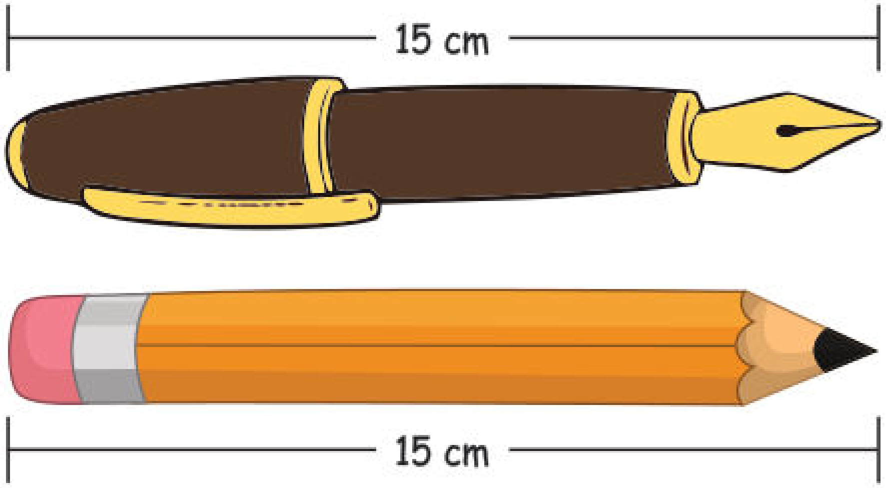
Example 4:

Compare the length of a book (20 centimeter) with that of an arm (45 centimeter) of a child.

As 20 is less than 45. So, the length of the book is **less** than that of an arm of the child.

Example 5:

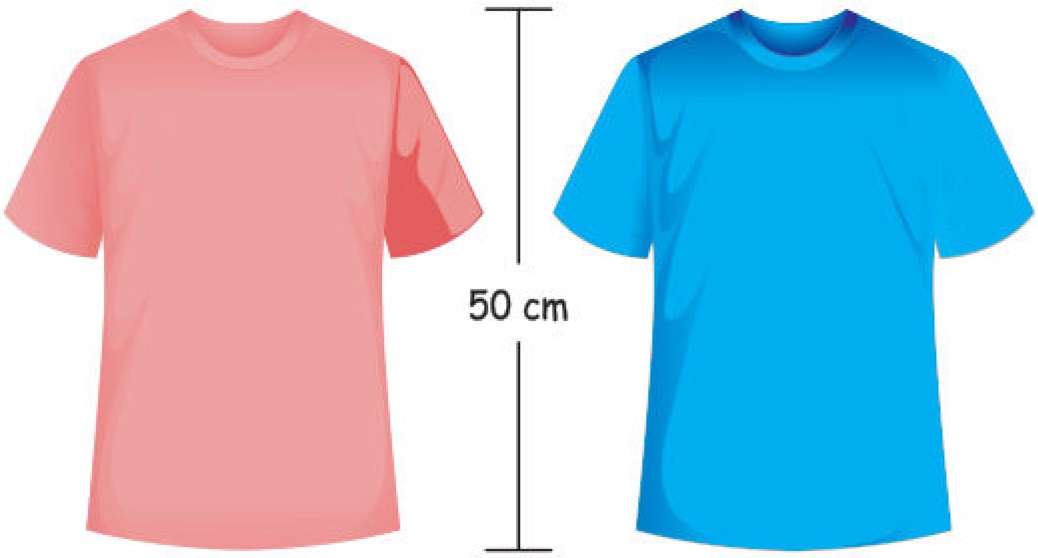
Compare the length of a pencil (15 centimeter) with that of a pen (15 centimeter) of a child.



As, 15 is equal to 15.Here, the length of the pencil is **equal** to that of a pen.

Example 6:

Compare the length of a red shirt (50 centimeter) with that of a blue shirt (50 centimeter).



As, 50 is equal to 50.So, the length of a white shirt (50 centimeter) is **equal** to that of a red shirt (50 centimeter).

Exercise 2

1. Which is longer, a pencil (15 cm) or a ruler (30 cm)?

   ______ is longer than ______

2. Is a bookshelf (2 m) longer or shorter than a bed (3 m)?

   ______ is shorter than ______

3. Compare the length of a football field (100 m) and a basketball court (28 m).

   ______ is shorter than ______

4. Which is longer, a shoe (25 cm) or a book (15 cm)

   ______ is longer than ______

5. Compare the length of a football field (100 m) and a cricket ground (100 m).

   Length of both football field and cricket ground is ______

6. Compare the length of a mobile 15cm with a pen 15cm.

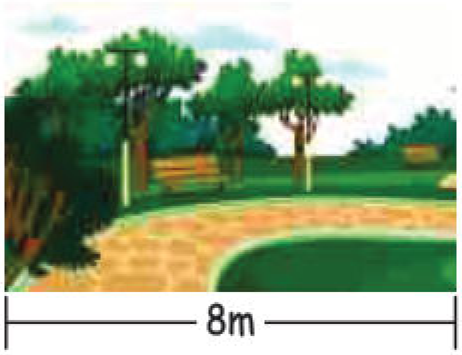
   Length of both mobile phone and pen is ______

Addition and Subtraction (Real life word problems):

We can add or subtract lengths of different objects in same units. in real-life. Here, some examples are given.

Example 1:

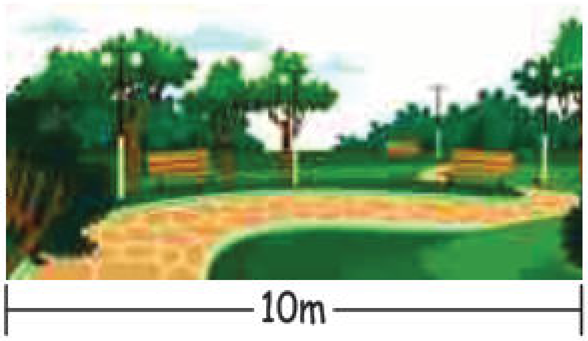
A path in a garden is 8 meters long. If we extend it 2 more meters, how long will it be?



Solution:

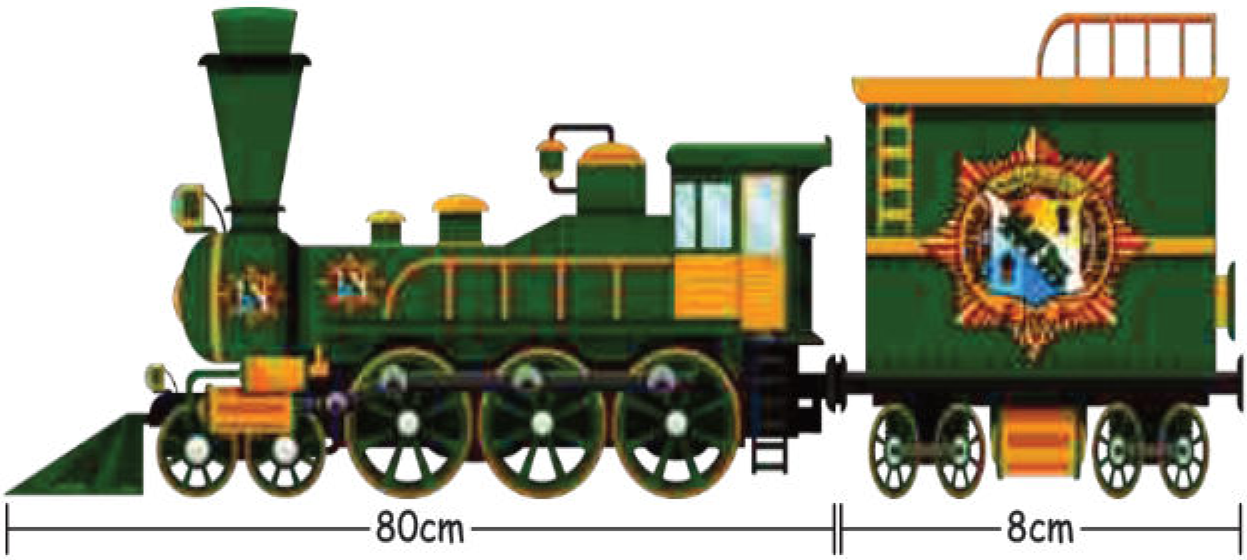
Length of path	=		8 m
Extra length	=	+	2 m
Total length	=		10m

Total length of path in a garden is 10m.



Example 2:

The length of toy train is 80cm and if one more coach of length 8cm is added into it then what will be the total length of the train?



Solution:

Length of a toy train	=		80 cm
Length of a coach	=	+	8 cm
Total length	=		88m

Example 3:

Hussain draws a line of 25cm long on a paper. Then, he removes the 10cm long part of the line of it. What is the length of remaining line?

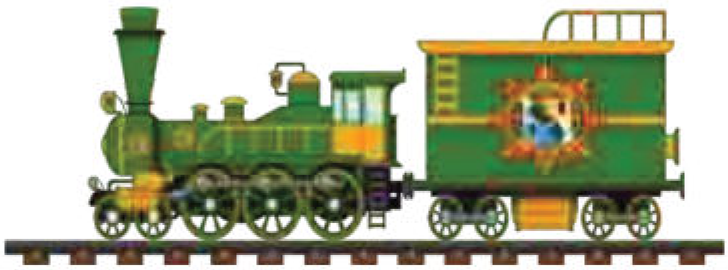
Solution:

Length of a line	=	25 cm
Length of removed part	=	- 10 cm
Length of remaining length	=	15 cm

Thus, the length of remaining line is 15 cm.

Example 4:

A toy train track is 18 meters long. If 5 meters of the track are removed, what is the length of the track now?



Solution:

Length of toy track	=	18 m
Length of removed track	=	- 5 m
Length of remaining track	=	13 m

Thus, the length of remaining track is 13 m.

Exercise 3

(1) A pencil is 15 centimeters long, and a ruler is 20 centimeters long. What is the total length of the pencil and the ruler?

Solution:

Length of a pencil = ___________

Length of a ruler = ___________

Total length = ___________

(2) A highway is 500 meters long, and a connector road is 250 meters long. What is the total length of the highway and the connector road?

Solution:

Length of highway = ____________

Length of a connector = ____________

Total length = ____________

(3) Ali runs 100 meters, and his friend runs 75 meters. How much farther did the athlete run?

Solution:

Ali runs = ____________

His friend runs = ____________

Ali runs further = ____________

(4) A dress is 90 centimeters long, and a tailor shortens it by 18 centimeters. How long is the dress now?

Solution:

Length of a dress = ____________

Tailor shortens = ____________

The current length of dress = ____________

(5) Ayesha has a 60cm ribbon. She used 40cm from it. What is the length of remaining piece of ribbon

Solution:

Total length of ribbon = ____________

Used ribbon = ____________

Remaining length of piece of ribbon = ____________

(6) Shahid's father is 185cm tall and Awais is 85cm shorter, than his father. What is Awais's height?

Solution:

Shahid's father height is = ____________

Awais is shorter than his father = ____________

Awais height is = ___________

Mass:

Have you ever thought? Why does a bag of books feel heavier than an empty one?

It is because of the mass or weight that bag of books than the empty one.

We weigh or measure mass of things to find how much heavy or light a thing is.

Standard Units of Mass:

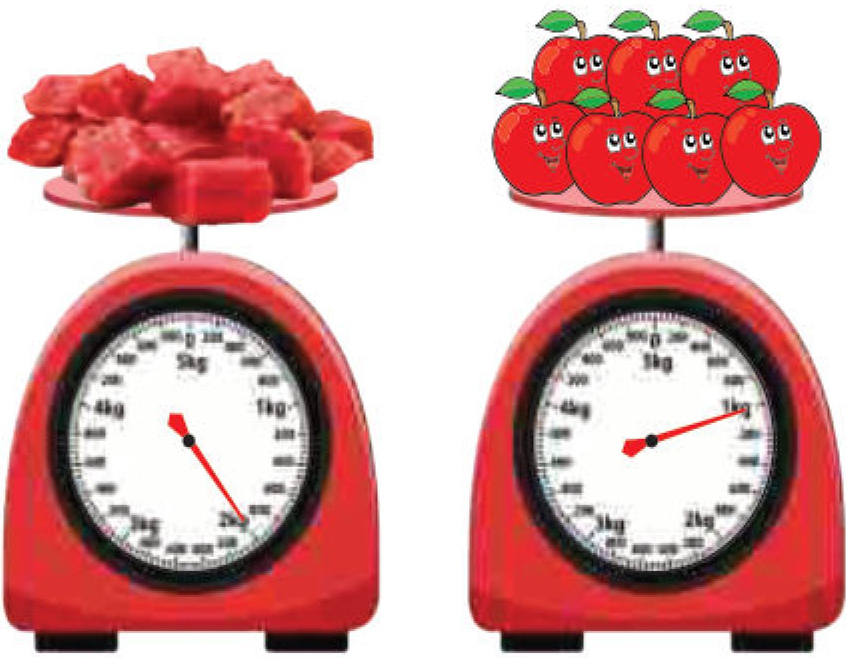
Do you know how much your favourite toy or book weighs? We can measure the mass of objects using standard units called kilograms (kg) and grams (gm).

Note: Here we use the term mass and weight interchangeably.

What is Kilogram?

Kilogram (Kg) is a standard unit of mass. We use kilograms to measure heavy things like a flour bag.

Example: The mass of apples is 1 kg.
The mass of meat is 2 kg.



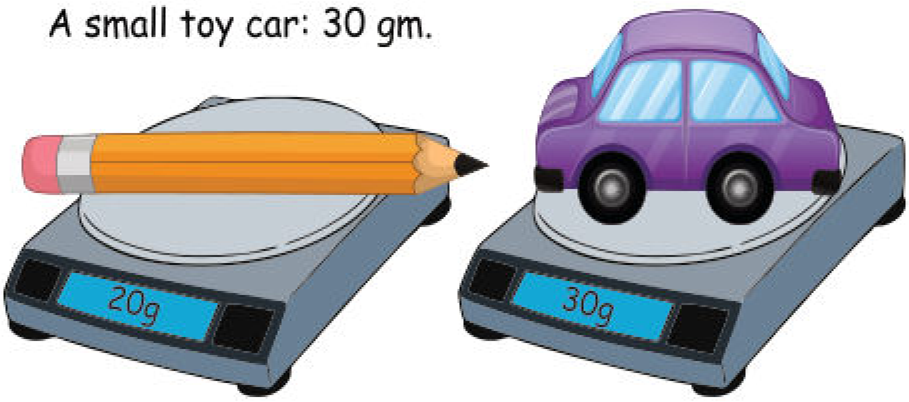
Activity

(i) Measure your weight in kilograms.
(ii) Measure the mass of your school bag in kilograms.

What is Gram?

Gram (gm) is a small unit of mass. We use grams to measure light things like a pencil or a small toy.

Example: Mass of pencil is 20 gm.
A small toy car: 30 gm.

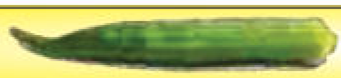


Remember:

- Kilograms (kg) are used for heavy things.
- Grams (gm) are used for light things.

Exercise 4

1. Write suitable unit to measure the weight.

(a)	An		weights in	gm	or	kg
(b)	A		weights in	gm	or	kg
(c)	A		weights in	gm	or	kg
(d)	A		weights in	gm	or	kg

2. Measure the weight of the following:

(a) Weight of a pencil in gm is ________

(b) Weight of a sharpener in gm is ________

(c) Weight of seven years old boy in kg is about ________

(d) Weight of a bag of tomatoes in kg is about ________

Comparison of mass of different objects:

We can compare the mass of different objects using standard units of mass (kilograms and grams). We can say if one object is heavier or lighter than another object.

Example 1:

Compare the mass of the toy truck (50gm) with that of a toy car (30gm).

As, 50 is greater than 30. So, the mass of the toy truck is greater than that of the toy car.

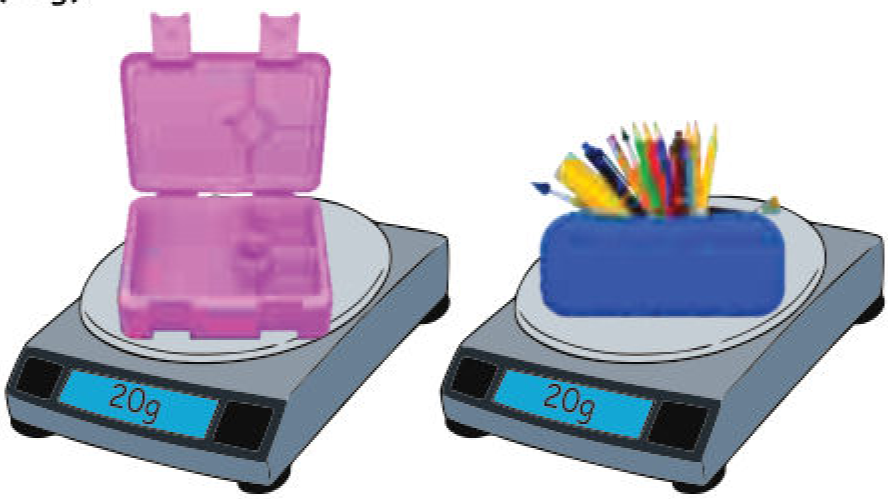
Example 2:

Compare the mass of a bottle of water (1kg) with that of a school bag (2kg).

As, 1 is less than 2. So, the mass of the bottle of water is less than that of the school bag.

Example 3:

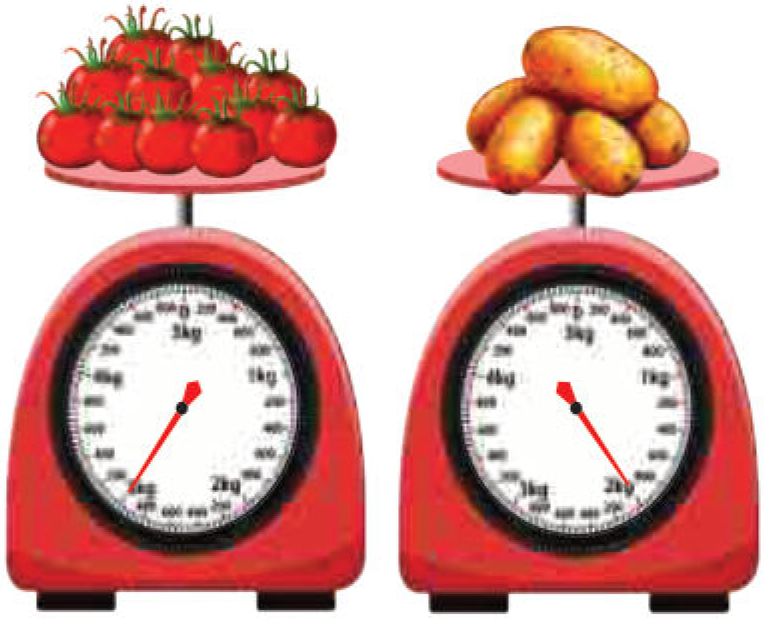
Compare the mass of a box of pencils (20gm) with that of empty lunch box (20g).



As, 20 is equal to 20. So, the mass of the box of pencils is equal to that of the empty lunch box.

Example 4:

Compare the mass of the tomatoes (3kg) with that of the potatoes (2kg).



As As, 3 is greater than 2. So, the mass of the tomatoes (3kg) is greater than with that of the potatoes (2kg).

Example 5:

Compare the mass of an apple (80g) with that of an orange (120g).

As 80 is less than 120. So, the mass of an apple is less than that of an orange.

Example 6:

Compare the mass of a sugar bag (5kg) with that of flour bag (5kg).

As, 5 is equal to 5.So, the mass of a sugar bag is equal to that of flour bag.

Exercise 5

1. A bag of apples weighs 3 kilograms, while a bag of arranges weighs 1 kilograms. Which has a greater mass.
   As, 3 is greater than 1
   So, __________ has greater mass.

2. A paperclip weighs 5 grams, while a pen weighs 20 grams. Which has a greater mass?
   As, 20 is greater than 5
   So, __________ has greater mass.

3. A small toy car weighs 50 grams, while a toy truck weighs 75 grams. Which has a lesser mass?
   As, 50 is less than 75
   So, _________ has lesser mass.

4. Ali's weight is 20 kg while his friend Amjad's weight is 24 kg. Who is lighter of them?
   As, 20 is less than 24
   So, the lighter of them is ___________.

5. Compare the weight of the book (500 grams) with that of the bag of sugar (500 grams).
   As, 500=500
   So, the weight of both book and sugar bag is ___________.

Addition and Subtraction of Mass:

Do you know how much sugar and flour are needed to make a delicious cake? Measuring mass (or weight) is an essential skill in daily life. In this section, we will learn how to add and subtract mass given in the same units to solve real-life word problems.

Example 1:

A student packs 100g of carrots and 80g of apples in her lunch box. What is the total mass of the lunch box?

Solution:

Mass of carrots	=	100g
Mass of apples	=	+ 80g
Total mass of lunch box	=	180g

Thus, the total mass of the lunch box is 180grams.

Example 2:

Asma buys 40kg of flour and 20kg of rice bags. What is the total mass of the both bags?

Solution:

Mass of flour bag	=	40kg
Mass of rice bag	=	+ 20kg
Total mass of both bags	=	60kg

Thus, the total mass of the flour and rice bags is 60 kilograms.

Example 3:

Ali bought 250g of strawberries. He ate 50 g of strawberries. How many gram of strawberries are left?

Solution:

Mass of total strawberries	=	250g
He ate strawberries	=	− 50g
Strawberries are left	=	200g

Thus, 200 grams of strawberries are left.

Example 4:

A bag of rice weighs 20kg. If 5kg of rice is taken out, what is the new mass of the bag?

Solution:

Mass of rice bag	=	20kg
Mass of rice taken	=	− 5kg
Mass of the new bag of rice	=	15kg

Thus, the mass of the bag is 15 kg.

Exercise 6

1. A bag of flour weighs 50kilograms. If 20 kilograms of sugar are added, what is the total weight of the bag?

Solution:

Flour weight = ___________

Sugar weight = ___________

Total weight of bag = ___________

2. A packet of coffee weighs 250 grams. If a packet of sugar weights sugar is 75 g, what is the total mass of the coffee and sugar together?

Solution:

Coffee weight = _________

Sugar weight = _________

Total weight = _________

3. A sack of potatoes weighs 15 kilograms. If 7 kilograms of potatoes are taken out, what is the remaining mass of the sack?

Solution:

Total weight of potatoes = ________

Taken out potatoes = ________

Wight of remaining potatoes = ________

4. A box of chocolates weighs 300 grams. If 250 grams of chocolates are eaten, what is the remaining mass of the chocolates?

Solution:

Weight of chocolate = _________

Chocolates are eaten = _________

Remaining weight of chocolate = _________

Capacity:

Capacity refers to the amount of liquid or substance that a container or object can hold.

For example, the jug can hold 8 glasses of water.

Here, the capacity of jug is 8 glasses.

Standard Units of Mass:

Standard units of capacity (liter and milliliter) are used to measure the amount of liquid or substance that a container can hold.

What is liter?

A liter (L) is a standard unit of volume or capacity, commonly used to measure the volume of liquids.

Measures used to measure the capacity of water is given as under:

Example:

Measure the capacity of given jug of water:

Now we use liter scale to measure the capacity of water in jug.

Here, the capacity of the jug of water is 1 liter.

Activity

Measure the capacity of the bottle of water you have in liter.

What is milliliter?

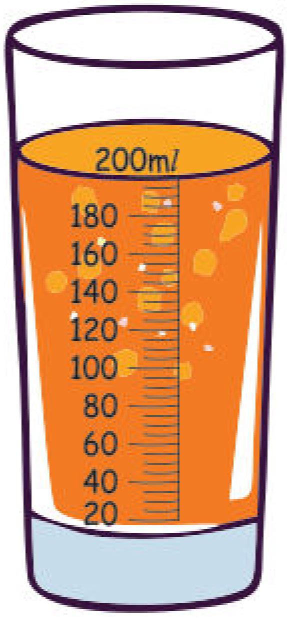
A milliliter (ml) is a tiny unit to measure liquids.

For instance, let's say you have a small water bottle that can hold 1000 tiny cups of water. Each tiny cup is like 1 milliliter (ml). So, the water bottle can hold 1000 milliliters (mL) of water.

Example:

Measure the capacity of the given glass of cold drink

Now we use milliliter scale to measure the capacity of cold water.



Here, the capacity of the glass of cold water is 200 ml.

Activity

Measure the capacity of glass of water in milliliter using syringe.

Comparison of capacity of different objects

Example 1:

Compare the capacity of the jug of water (2 liter) with that of the packet of juice (1 liter).

As, 2 is greater than 1. So, the capacity of Jug of water is greater than that of the packet of juice.

Example 2:

Compare the capacity of the glass of soda (100 milliliter) with that of the medicine dropper (20 milliliter).

As, 100 is greater than 20. So, the capacity of the glass of soda is greater than that of the medicine dropper.

Example 3:

Compare the capacity of fuel tank (20 liter) with that of the another large fuel tank (50 liter).

As, 20 is less than 50.So, the capacity of fuel tank is less than that of the large fuel tank.

Example 4:

Compare the capacity of cup of tea (50 milliliters) with that of the large cup of tea (100 milliliters).

As, 50 is less than 100.So, the capacity of cup of tea is less than that of the another large cup of tea.

Example 5:

Compare the capacity of container of oil (500 liter) with that of the container of water (500 liters).

As, 500 is equal to 500.So, the capacity of container of oil is equal to that of the container of water.

Example 6:

Compare the capacity of perfume bottle (50 milliliters) with that of the perfume bottle (50 milliliters).

As, 50 is equal to 50.So, the capacity of perfume bottle is equal to that of the perfume bottle.

Exercise 7

1. A packet of juice has a capacity of 200 milliliters. A milk packet has a capacity of 250 milliliters. Which has a greater capacity?
   As, 250 is greater than 200
   So, __________ has greater capacity.

2. A fuel tank A has a capacity of 60 liters. Another fuel tank B has a capacity of 65 liters. Which has a lesser capacity?
   As, 60 is less than 65
   So, __________ has lesser capacity.

3. A hand wash bottle has a capacity of 250 milliliters. A shampoo bottle has a capacity of 500 milliliters. Which has a greater capacity?
   As, 250 is less than 500.
   So, ___________has lesser capacity.

4. Compare the capacity of a water cooler 4 liter with that of the fuel gallon 4 liter.
   As, 4 is equal to 4.
   So, the capacity of water cooler and fuel gallon is ____________

5. A water tank has a capacity of 700 liters. A fuel tank has a capacity of 600 liters. Which has a greater capacity?
   As, 700 is greater than 600
   So, __________ has greater capacity.

6. Compare the capacity of a bottle of soap 100 milliliter with that of the bottle of perfume 100 milliliter.
   As, 100 is equal to 100.
   So, the capacity of bottle of soap and bottle of perfume is

Addition and Subtraction:

We can add or subtract capacity of different objects in same units in real-life. Here, some examples are given.

Example 1:

A juice manufacturer has 300 liters of apple juice and 50 liters of orange juice. How much juice does the manufacturer have in total?

Solution:

Capacity of apple juice	=	300L
Capacity of orange juice	=	+ 50L
Total capacity of juices	=	350

Thus, the total capacity of juices is 350 liter.

Example 2:

A pharmacist has 250 mL of a certain medicine in a bottle. She needs to add 100 mL more of the medicine to the bottle. How much medicine will the bottle contain after the addition?

Solution:

Volume of medicine in a bottle	=	250mL
Volume of extra medicine of bottle	=	+ 100mL
Total volume of medicine of bottle	=	350mL

Thus, the total capacity of medicine of bottle is 350 milliliters.

Example 3:

A water tank can hold 500 liters of water. If 50 liters of water are poured out, how much water is left in the tank?

Solution:

Capacity of water in tanker	=	500L
Capacity of water poured out	=	− 50L
Remaining water in the tanker	=	450L

Thus, the total capacity of water is 450 liter.

Exercise 8

1. A packet of juices has a capacity of 200 milliliters. Another packet of juice is of 100 milliliters, what is the total capacity of juice and packets?

Solution:

Capacity of a juice packet = __________
Capacity of another packet = __________
Total capacity of juice = __________

2. A water tank has a capacity of 800 liters and a fuel tank has the capacity of 500 liters, what is the total capacity of both tanks

Solution:

Capacity of water tank = __________
Capacity of fuel tank = __________
Total capacity of both tanks = __________

3. A gas cylinder has a capacity of 45 liters. If 15 liters of gas are used for cooking, what is the remaining capacity?

Solution:

Total capacity of gas Cylinder = ________

Used capacity of Cylinder = __________

Remaining capacity of Cylinder = __________

4. A perfume bottle has a capacity of 100 milliliters. If 50 milliliters of perfume bottle are used, what is the remaining capacity of perfume?

Solution:

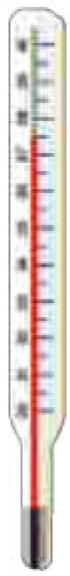
Total capacity of perfume bottle = ________

Used capacity of Perfume bottle = ________

Remaining capacity of Perfume bottle = ________

Temperature:

Have you ever felt hot or cold before?
You definitely feel hot in summer when you go outside school or home, and similarly, you feel cold in winter when you do not wear a sweater.

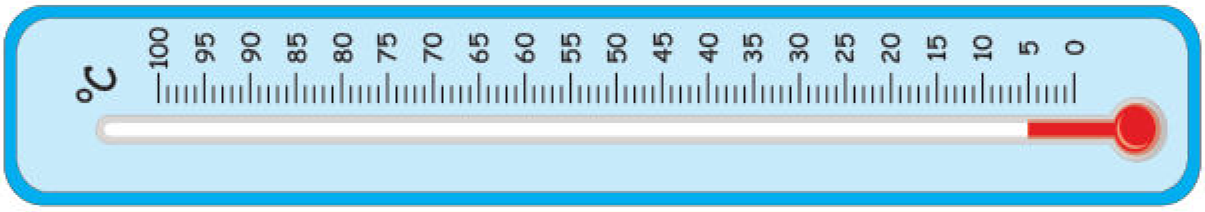


The temperature is a measure of how hot or cold something is. It is measured in unit called degree Centigrade (°C). Thermometer is used to know how much degrees Centigrade of body is.

Reading and Writing of Temperature:

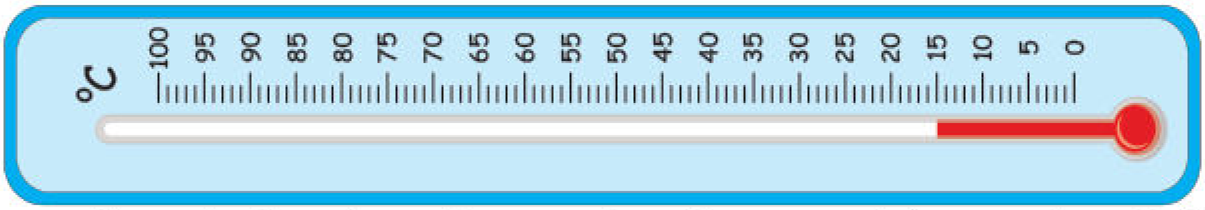
Here, we read and note the temperature on Celsius scale (°C) using thermometer.

Example 1: Read the temperature in the given thermometer



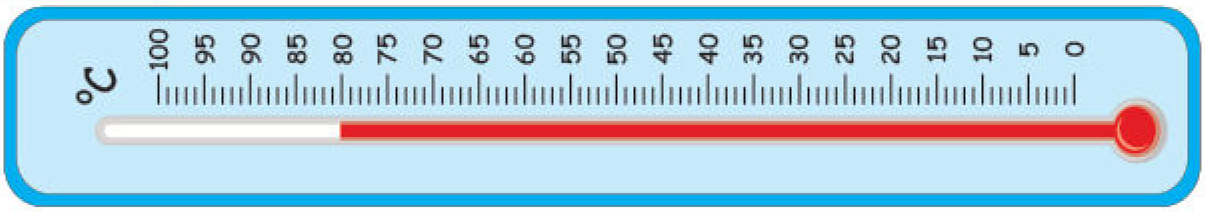
Here, in the given thermometer the nearest temperature is 5 degree Centigrade.

Example 2: Read the temperature in the given thermometer



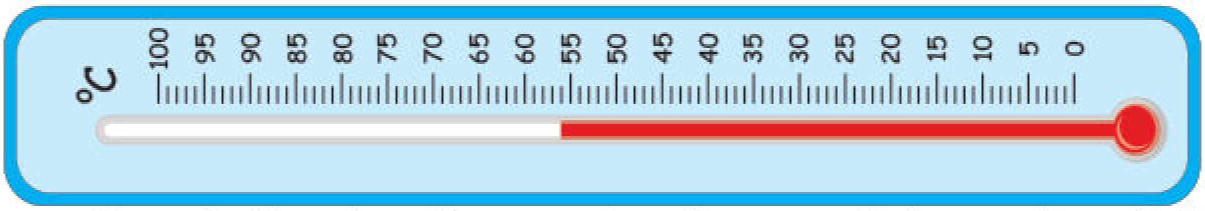
Here, in the given thermometer the nearest temperature is 15 degree Centigrade.

Example 3: Read the temperature in the given thermometer



Here, in the given thermometer the nearest temperature is 80 degree Centigrade.

Example 4: Read the temperature in the given thermometer

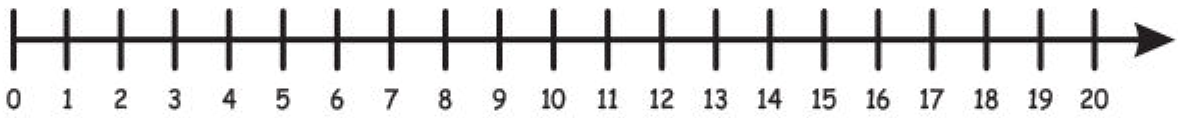


Here, in the given thermometer the nearest temperature is 56 degree Centigrade.

Relating temperature scale to number line

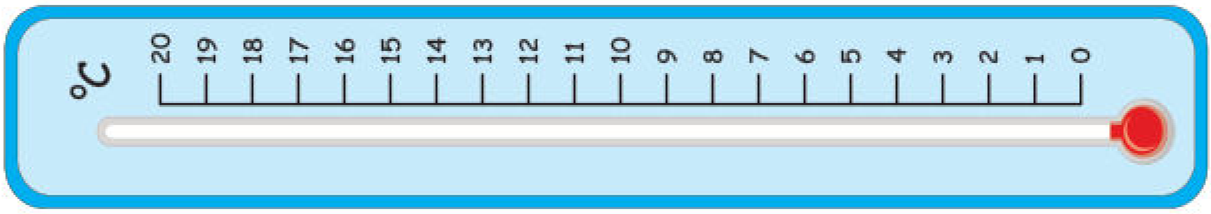
We use a number line to show numbers in order from smallest to largest. We can think of the temperature scale like a number line, too!

Number Line



The above figure represents the Number line from 0 to onward.

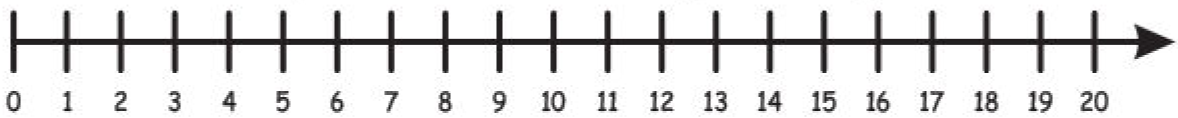
Similarly we can treat temperature scale as a number line. For example, thermometer scale can work as a number line too.



On a number line, we have numbers like 0, 5, 10,20 and so on. On the temperature scale, we have temperatures like 0°C, 5°C, 10°C, 20°C and so on.

Now the reading obtained from thermometer can be represented on Number line.

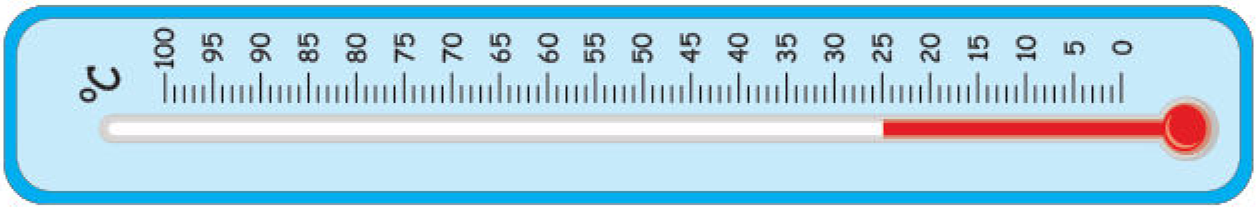
Temperature Scale into Degree Centigrade



Just like how numbers get bigger as we move to the right on a number line, temperatures increase as we move to the right on the temperature scale. And just like how numbers get smaller as we move to the left on a number line, temperatures decrease as we move to the left on the temperature scale

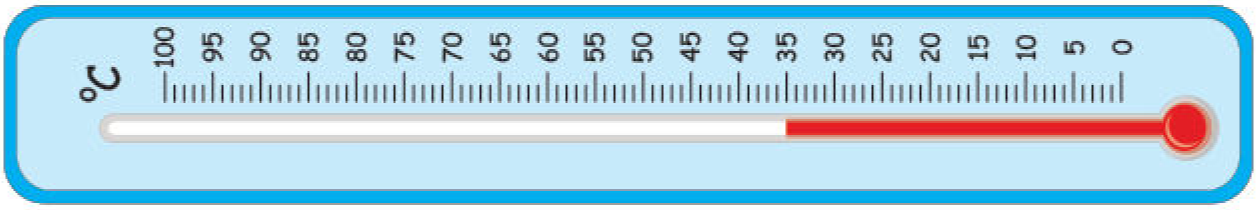
Exercise 9

1. Read the temperature in the given thermometer



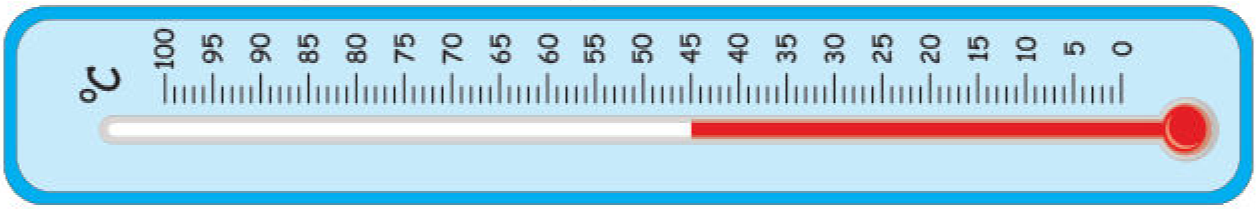
The temperature of the thermometer is __________.

2. Read the temperature in the given thermometer



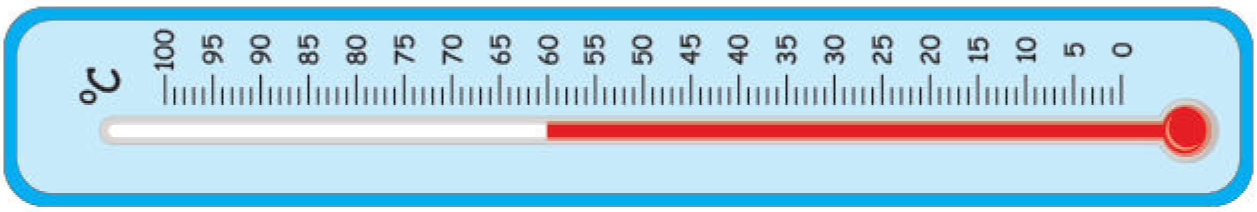
The temperature of the thermometer is __________.

3. Read the temperature in the given thermometer



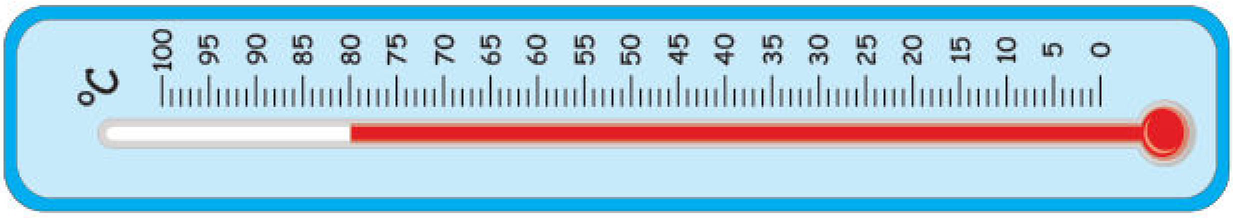
The temperature of the thermometer is __________.

4. Read the temperature in the given thermometer



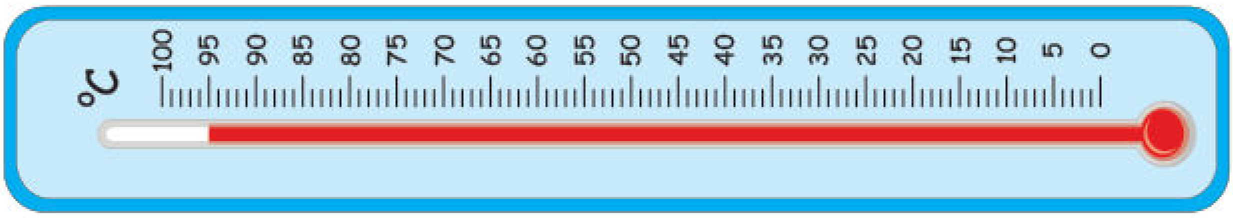
The temperature of the thermometer__________.

5. Read the temperature in the given thermometer and show it on number line.



The temperature of the thermometer_________.

6. Read the temperature in the given thermometer and show it on number line.



The temperature of the thermometer_________.

Unit 7

TIME

Student Learning Outcomes:

At the end of the chapters, Students will be able to:

- ✓ Show time in hours and minutes on an analogue clock.
- ✓ (duration) of time (for instance to estimate/give a rough calculation of the time taken by particular events or tasks)
- ✓ Use calendar to find a particular day/date (of a week/month) in real life situation.